

Rajat Savdekar

rajatsavdekar@gmail.com • linkedin.com/in/rajatsavdekar • [GitHub](https://github.com) • rajatsavdekar.dev • (213) 574-8894 • LA, CA

EDUCATION

University of Southern California, Los Angeles, CA Aug 2024 – May 2026
Master of Science in Computer Science
GPA: 3.63
Coursework: Algorithms, Database Systems, Machine Learning, Web Technology, Distributed Systems

University of Mumbai, Mumbai, India Aug 2019 – May 2023
Bachelor of Engineering in Computer Engineering
CGPA: 9.4/10
Coursework: Operating Systems, Computer Networks, Distributed Systems, Natural Language Processing

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL, React, Next.js, HTML, Tailwind CSS

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets, gRPC

Cloud & Infrastructure: AWS (EC2, S3, CloudWatch), Docker, Terraform, GitHub Actions, CI/CD

EXPERIENCE

Software Developer Aug 2026 – Present
Easley-Dunn Productions, Inc. – Mediverse Torrance, CA

- Architected a **multithreaded** state synchronization engine across 6 concurrent modules, implementing **object pooling** and **lock-free concurrency** in **C#** to eliminate **garbage collection (GC)** overhead and race conditions, guaranteeing deterministic sub-15ms execution cycles.
- Engineered a low-latency asynchronous inference pipeline for a **local quantized LLM**, managing concurrent data streaming and memory allocation within a strict 4 GB RAM limit to reduce processing latency from 22 to 4 minutes.

Software Engineer May 2023 – May 2024
Jalgaon Fruit Sales Cooperative Ltd (JFSS) Jalgaon, India

- Doubled peak harvest write throughput by decomposing a **legacy PHP monolith** into **Python Flask microservices** on **PostgreSQL**, eliminating MyISAM table-lock freezes and cutting query latency 40%.
- Containerized services with **Docker**, deployed to **AWS EC2** via **GitHub Actions CI/CD**, and instrumented with **CloudWatch** monitoring and **pytest** integration tests enforcing 80% coverage.
- Built an asynchronous document processing pipeline handling 1,000+ daily transactions, implementing **exponential backoff** on rate limits and **Redis caching** to support a **GPT-4** automated reconciliation service.
- Architected a high-throughput **semantic search service** over 150K+ procurement records using **OpenAI embeddings** and **ChromaDB**, cutting average data retrieval time from 3 minutes to under 20 seconds.

PROJECTS

MockPad – Real-Time Collaborative Code Editor [Live] [GitHub]
TypeScript, Node.js, React, WebSockets, Yjs CRDTs, OpenAI Whisper, MongoDB, LevelDB Aug 2026

- Scaled to **10–20 daily active users** on a \$6/month droplet by combining a Yjs CRDT sync layer over **WebSockets** with an LRU eviction policy offloading idle rooms to **LevelDB**, sustaining **100% uptime** under concurrent load.
- Shipped **session replay** with deterministic playback and a **React + TypeScript** frontend on **Monaco Editor** delivering a free, self-hostable alternative to CoderPad for live interviews.

DevStack – AI-Native Developer Infrastructure Platform [GitHub]
TypeScript, Python, Node.js, PostgreSQL, Redis, TimescaleDB, Docker, FastAPI Aug 2026

- Architected 8 **microservices** behind an **API gateway** with **JWT**, **RBAC**, sub-5ms feature flag evaluation via **Redis pub/sub** and **WebSocket** push, and a workflow engine using **event sourcing**.
- Distributed isolated **Docker sandboxes** for scalable workload execution using **PostgreSQL SKIP LOCKED** queues with lease-based crash recovery, supporting an **LLM-as-judge** evaluation engine.
- Developed an **event-driven webhook processor** that triggers on GitHub CI failures, retrieves context via a vector index, and automatically opens fix PRs with structured patch outputs.

Jev Tools – Real-Time Voice Command Console [Demo] [GitHub]
TypeScript, Next.js, SQLite, TypeSafe APIs, Event-Driven Architecture Sep 2026

- Engineered a real-time speculative execution engine processing concurrent data streams, paired with an append-only **SQLite** ledger to guarantee zero data loss and strict auditability for every asynchronous routing decision.
- Built an **automated telemetry calibration harness** utilizing **algorithmic perturbations** to lock execution thresholds, enforcing **deterministic request routing** and reducing calibration error to 0.05.

RESEARCH & HACKATHONS

Publications. *Texture Feature Analysis* – MATLAB tool cited by 10+ papers in IEEE and Springer volumes. [GitHub]

Hackathons. Competed in 6+ events including **OpenAI Build Week**, **CALL-E 2026 Reddit Devvit** [Demo].